

Shivam Shukla

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EDUCATION

Vellore Institute of Technology

Integrated M.Tech. in Computer Science and Engineering (Data Science) | **CGPA: 8.64/10**

Bhopal, India

Expected 2026

AKLANK Day Boarding School

Senior Secondary (Class XII) | **88.2%**

Kota, Rajasthan

2020

TECHNICAL SKILLS

Languages: Python, SQL, C++

AI & Deep Learning: Machine Learning, Deep Learning (CNN, RNN, GRU, BiLSTM), NLP, RAG, Vector Search

Frameworks & Libraries: TensorFlow, Keras, Flask, Scikit-learn, NumPy, Pandas, Matplotlib, SentenceTransformers, NLTK

Backend & Databases: Bootstrap, Jinja2, MySQL, SQLite, SQLAlchemy (ORM)

Core Competencies: Data Structures & Algorithms (DSA), Data Preprocessing & EDA, Model Building, REST APIs, Model Deployment, Git

PROJECTS

Institute Placement Management System | *Flask, SQLite, SQLAlchemy, Jinja2*

 [Live Demo](#)

- Developed a full-stack recruitment portal with **Role-Based Access Control (RBAC)** for Admins, Companies, and Students.
- Built a programmatic database schema managing complex relationships for scalable data handling of multiple concurrent student applications and job drives.
- Leveraged Jinja2 templating for server-side rendering, ensuring a secure and efficient user workflow.

Amazon E-commerce Agent | *Python, Flask, ChromaDB, SentenceTransformers*

 [Source](#)

- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline for semantic product search and recommendation.
- Built an end-to-end ML workflow: raw data preprocessing → vector embeddings → similarity retrieval → LLM response.
- Deployed a responsive Flask interface to handle natural language user queries for product discovery.

Master Thesis Research: Network Intrusion Detection | *TensorFlow, Keras, Deep Learning*

 [Draft Paper](#)

- Engineered an attention-augmented GRU-BiLSTM hybrid deep learning model to capture detailed temporal context and improve explainability in traditional black-box NIDS.
- Paper Submitted:** "Network Intrusion Detection System Using Hybrid Approach of GRU-BiLSTM Attention-Based Deep Learning Model" to the 6th International Conference on Intelligent Technologies (ICIT 2026).
- Evaluated model robustness across diverse network environments using **UNSW-NB15**, **CIC-IDS2017**, and **TON-IoT** datasets, achieving a peak detection accuracy of **99%**.

ACHIEVEMENTS & CERTIFICATIONS

Competitive Programming: Solved **450+** problems on LeetCode (Rating: 1602, Top ~23%) and **500+** problems on GeeksforGeeks.

Contests: Actively participate in algorithmic contests with a rating of **1176** on **Codeforces**.

Certifications: Applied Machine Learning in Python (Univ. of Michigan, Coursera) | Data Analysis with Python (IBM, Coursera).

INTERESTS AND HOBBIES

Technical: Algorithmic Problem Solving, Application Deployment, Exploring Emerging AI & Web Technologies.

Recreational: Playing Cricket, Outdoor Team Sports.